

The theoretical research of the rate of chemical reactions

Introduction

Chemistry is all about reactions and creating. During each reaction, despite of the differences of reactants and products, the mechanism of different reactions are also very different from each other. In organic, there are different type of reactions and different mechanisms for nearly every type of organic functional groups. Different mechanisms may also lead to different in the speed of the reaction. This is called as the rate of reaction. There are various aspects that can affect the rate of a reaction. This includes pressure, concentration of reactants, and temperature. Therefore, in this experiment, I choose one of the aspects, which is temperature, to determine its relationship to the rate of reaction. The measure is based on the reaction between *sodium thiosulfate* and *hydrochloric acid*.

Background Information

The rate of reaction is the change of the concentration of reactants or products. It also means the decreasing of the concentration of reactants or the increasing of the concentration of products.

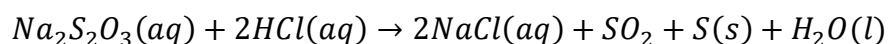
In theory, the rate of reaction will increase with the increase of temperature. This is because temperature is directly related to the kinetic energy of the particle. **The Kelvin temperature has a positive relationship to the kinetic energy a particle contains.** When temperature increases, the average kinetic energy of particles increase as well. With more speed, the particles will contain higher energy which will be equal or larger than the activation energy. This is the minimum amount of energy that the particles must contain in order to start a successful reaction. More particles have ability to engage in successful reaction, therefore, increasing the collision frequency. Since chemical reactions are all about colliding, an increasing in the collision frequency will therefore leads to an increase of the rate of reaction.

Research Question

As written in the background information, the rate of reaction has a positive relationship to the absolute temperature the particles contain. Therefore, this experiment is going to verify such relationship between the rate of reaction and the absolute temperature.

Methodology

Reaction



Chemicals and Apparatus

1. 60 cm^3 of 0.10 mol dm^{-3} of aqueous $Na_2S_2O_3$
2. 60 cm^3 of 0.10 mol dm^{-3} of aqueous HCl
3. two test tubes
4. a beaker with a cross labeled at the bottom
5. a timer
6. a water bath
7. a thermometer
8. a measuring cylinder

Procedures

1. Label the two test tubes with *A* and *B*.
2. Make the beaker with a cross at the bottom together with the two test tubes into the water bath.
3. Set up the temperature of the water bath to $30\text{ }^\circ\text{C}$.
4. Place 10 cm^3 of $Na_2S_2O_3$ and HCl into tube *A* and tube *B* separately.
5. Wait until they both reach thermal equilibrium.
6. Mix the two reactants into the beaker and start the timer immediately.
7. Record the time when the cross disappear completely and the temperature when this phenomenon occurs.
8. Repeat the process from 3 to 7 until getting five results at five different temperatures. During each time, add the temperature of the water tube for $5\text{ }^\circ\text{C}$.

Variables

Independent Variable Temperature (<i>K</i>)	Dependent Variable Rate of Reaction (<i>s</i>)	Constant Variable $[Na_2S_2O_3]$ (mol m^{-3})	Constant Variable $[HCl]$ (mol m^{-3})	Constant Variable Volume of $Na_2S_2O_3$ Used (m^3)	Constant Variable Volume of HCl Used (m^3)
303	/	0.10	0.10	0.01	0.01
308	/	0.10	0.10	0.01	0.01
313	/	0.10	0.10	0.01	0.01
318	/	0.10	0.10	0.01	0.01
323	/	0.10	0.10	0.01	0.01

Set up Diagrams



Figure 1: a set up diagram taken for the experiment

As Figure 1 has shown, the experiment is conducted in a water bath. The two solution ahead are 60 cm^3 of 0.10 mol dm^{-3} of aqueous $\text{Na}_2\text{S}_2\text{O}_3$ and 60 cm^3 of 0.10 mol dm^{-3} of aqueous HCl separately. During each trial of the experiment, a specific amount of the solutions will be transferred into two test tubes. They will be hold in a beaker with water in the water bath.

Safety, Ethical, and Environmental Issues

In this experiment, a water bath is used. There is something that need to be aware of when using such equipment during the experiment. The water bath has to set a temperature that needs it to heat to before heating the water. The temperature can only be larger than the current water temperature. In other words, it is impossible for a water bath to decrease temperature. The only way for it to do so is to wait or change a cooler water. In order to protect the equipment, the temperature set must not be lower than the current water temperature.

Data Analysis

Raw Data

Qualitative Data

Before the reaction, the two solutions are all colorless. After mixing them together, the solution will turn to blurry. This is due to the precipitation of Sulfur. Therefore, by determining the time when the solution become not transparent enough to see the mark on the beaker, it can show the time for the reaction to complete. The rate of reaction means the rate of the increasing of concentration of products. If the amount of products formed is same but with different time, different rate of reaction can therefore be shown.

Quantitative Raw Data

Table 1: time for reaction to complete under different temperature

Temperature (K)	Trial 1 (s)	Trial 2 (s)	Trial 3 (s)
303	86.28 ± 0.01	85.66 ± 0.01	86.55 ± 0.01
308	51.53 ± 0.01	52.36 ± 0.01	51.23 ± 0.01
313	50.66 ± 0.01	50.36 ± 0.01	49.64 ± 0.01
318	47.04 ± 0.01	48.12 ± 0.01	47.56 ± 0.01
323	28.94 ± 0.01	26.67 ± 0.01	27.88 ± 0.01

Processed Data

Make the temperature of 303 K as an example to process the data.

There are 3 different trials of data under a same temperature condition. In the case of 303 K, the three different reaction times are 86.28, 85.66, and 86.55. The unit for these three data is seconds. Therefore, an average reaction time can be determined:

$$\begin{aligned}
 \bar{t}_1 &= \frac{86.28 + 85.66 + 86.55}{3} \\
 &= 86.1633 \\
 &\approx 86.16 \text{ s}
 \end{aligned}$$

The uncertainty has been added up. Therefore, the uncertainty for $\bar{t}_1$ is:

$$\begin{aligned}
 \Delta \bar{t}_1 &= \frac{t_{\max} - t_{\min}}{2} \\
 &= \frac{86.55 - 85.66}{2} \\
 &= 0.445 \\
 &\approx 0.4
 \end{aligned}$$

Hence, the processed data for the reaction time under an absolute temperature at 303 K has been determined:

$$\bar{t}_1 = (86.2 \pm 0.4) \text{ s}$$

The other four different data can also be processed through the same procedures listed above. Therefore, a table can be created for the processed data:

Table 2: processed data for the experiment

Temperature(K)	Time for the Reaction (s)	Uncertainty (s)
303	86.2	0.4
308	51.7	0.6

313	50.2	0.5
318	47.6	0.5
323	29	1

As it has been predicted and proved theoretically in the background information part, the rate of reaction has a positive relationship with the increasing of absolute temperature. The unit for rate of reaction is s^{-1} . Hence, the data of time of reaction needs to be further processed into its reciprocal.

For its uncertainty, it can be calculated through the method of percentage uncertainty. Making the data of temperature of 303 K as an example again:

$$\begin{aligned}\frac{1}{\bar{t}_1} &= \frac{1}{86.2} \\ &= 0.011600 \\ &\approx 0.0116 \text{ s}^{-1}\end{aligned}$$

The percentage of the original time for the reaction is:

$$\begin{aligned}\Delta \bar{t}_1 \% &= \frac{0.4}{86.2} \\ &= 4.6 \times 10^{-3}\end{aligned}$$

Hence, the absolute uncertainty for the reciprocal of $\bar{t}_1$ is:

$$\begin{aligned}\Delta r_1 &= r \times \Delta \bar{t}_1 \% \\ &= 0.0116 \times 4.6 \times 10^{-3} \\ &= 5.336 \times 10^{-5} \\ &\approx 5 \times 10^{-5} \text{ s}^{-1}\end{aligned}$$

As a result,

$$r_1 = (0.0116 \pm 0.0001) \text{ s}^{-1}$$

By following the procedures above, the other four data left can also be processed. The table is illustrated below:

Table 3: table with absolute temperature and the rate of reaction

Absolute Temperature(K)	Rate of Reaction (s^{-1})	Uncertainty (s^{-1})
303	0.0116	0.0001
308	0.0193	0.0002
313	0.0199	0.0002
318	0.0210	0.0002
323	0.036	0.001

The entire processed data are loaded into Logger Pro to find the trend of the relationship between the absolute temperature and the rate of the reaction. This is shown below through Figure 2:

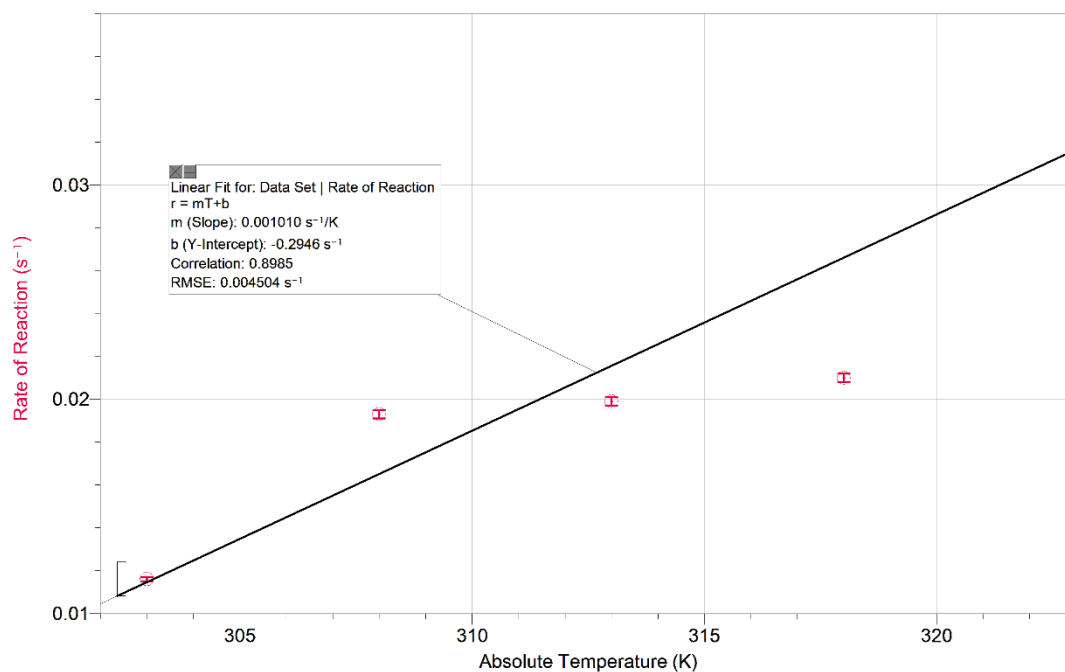


Figure 2: a general trend for the relationship between the absolute temperature and the rate of reaction

Conclusion

The aim of the experiment is to verify the relationship between the absolute temperature of a reaction and its rate to complete the reaction. According to the chemical collision theory, the absolute temperature has a positive relationship to the rate of the reaction. In other words, when the absolute temperature increases, the rate of reaction increases as well. Through Figure 2, the positive relationship can be shown. As the absolute temperature increases from 303 K to 323 K, the rate of reaction increases from 0.0116 s^{-1} to 0.03593 s^{-1} . Such relationship is thus being verified.

Evaluation

Although the aim of the experiment has been successfully accomplished, there are also some causes that might result in the differences between the theoretical curve and the experimental curve. In theory, the line should be a positive slope line that passes through the origin. However, in the experiment, the line has a y-intercept which is -0.2946 . This is result from systematic errors.

For some possible reasons for systematic errors, one might be the temperature of each trial is actually lower than the expected value. This is because, the water bath cannot maintain the water temperature for a long time. Therefore, when moving the tubes into the water bath, the temperature inside has already gotten down. Moreover, since I do this experiment in Winter, the room temperature is also low. Thus, the heat in the water bath will leave quickly. For a lower temperature, the rate of reaction will be decreased. As a result, the time taken is increased for the observation to occur. Smaller value of reciprocal of time is thus being determined. The whole experimental line is therefore being shifted downward.

This error can be solved by improving the environment of the experiment. If the experiment is conducted in Winter, the room temperature should also be adjusted to standard condition by air conditioner. This should also be adjusted in Summer when the room temperature is especially hot. Moreover, the water bath should be placed into an isolated system with a thermometer to make sure the temperature is exactly around the test value.

Moreover, there might also be some random errors that lead to the uneven spread of the experimental data. One cause might be the speed of stir during the experiment. In the experiment, I stir the solution manually; thus, the speed varies from each trials. Since the speed of stirring can affect the rate of reaction, thus the rate of reaction will vary around the best-fit line just as what has been shown in Figure 2.

A simple way that can solve the problem is to use a magnetic stirrer. Such device can

control the speed of stir using magnets which is a mechanical way. By doing so, the speed of stirring can be controlled and thus make such variable to stay constant during the experiment.

At last, I only verify the general trend of the relationship between absolute temperature and the rate of the reaction. However, the exact mathematic equation between such two variables are not being found and investigated. Such topic is also worth to be investigated on and can be considered as the topic for the future investigations.